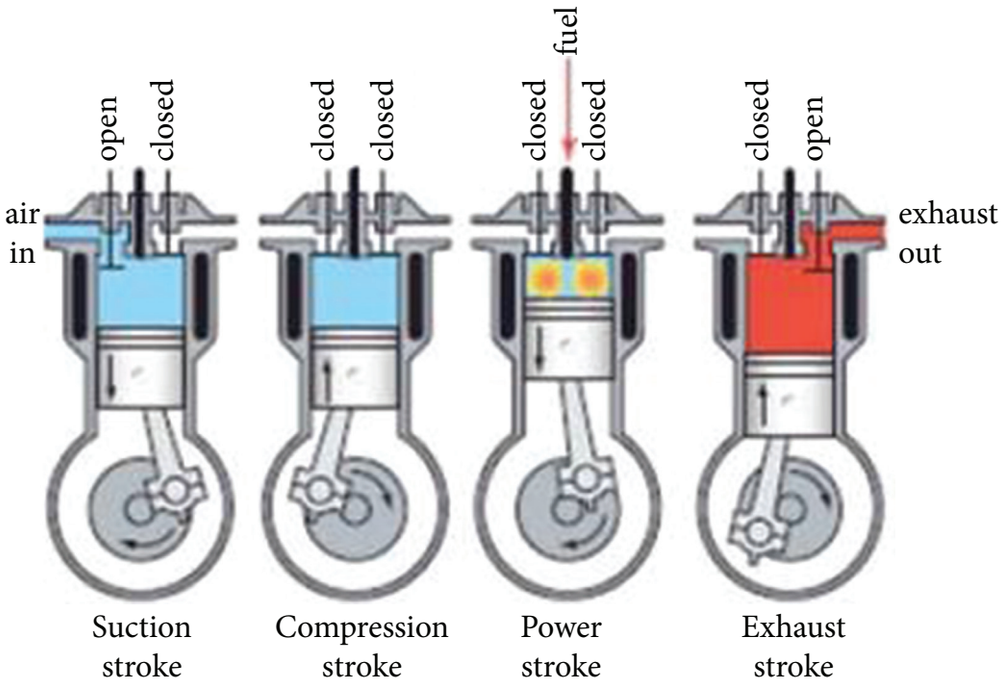


Figure 5.7(b) Working of Four Stroke Diesel Engine

crankshaft and on another side a pulley is attached to it. Inlet valve, exhaust valve and fuel injector are located on the engine head. The inlet valve will allow the air inside the cylinder, the fuel injector will inject the fuel and the exhaust valve will sent the burnt gases to the atmosphere. The valves are operated by the camshaft. The camshaft is driven by crankshaft through timing gear.

The following sequence of events will occur continuously for the engine operation. The figure shows the sequence of events

- ❒ Air alone induced inside the cylinder
- ❒ Air alone compressed
- ❒ Finely automised diesel fuel is injected onto the compressed air followed by combustion
- ❒ Burnt gases are let out from the cylinder.

The above sequence of operation forms a cycle and for each operation, a piston stroke is required. Thus for four operation, four strokes are required and hence it is called as four stroke engine. The four strokes are namely called

i. Suction Stroke
Ii. Compression Stroke
Iii. Power Stroke
Iv. Exhaust Stroke

i) Suction Stroke

In this stroke, the piston moves down from the top dead centre towards the bottom dead centre. The inlet valve opens and the air is drawn into the cylinder due to the vacuum created inside the cylinder. To fill the vacuum, the air is entered inside the cylinder. The inlet valve closes at the end of the stroke and the exhaust valve remains closed during this stroke.

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